

# A small mathematical notebook

## Derivatives, integrals, and a useful identity

2026-10-06

This is a sample document for a personal notebook. The source is Markdown with LaTeX math; the PDF is typeset by Typst.

### 1. A derivative from first principles

Let  $f(x) = x^2$ . Starting with the definition,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} (2x+h) = 2x.$$

The derivative describes the local rate of change. At  $x = 3$ , the slope is  $f'(3) = 6$ .

### 2. An integral

The corresponding antiderivative gives

$$\int_a^b x^2 \, dx = \frac{b^3 - a^3}{3}.$$

For example,  $\int_0^1 x^2 \, dx = 1/3$ .

### 3. A useful identity

For vectors  $u, v \in \mathbb{R}^n$ , the Cauchy-Schwarz inequality states that

$$\left| \sum_{i=1}^n u_i v_i \right| \leq \sqrt{\sum_{i=1}^n u_i^2} \sqrt{\sum_{i=1}^n v_i^2}.$$

Equality holds when the vectors are linearly dependent.

### 4. Keep a reproducible note

1. State the question and the assumptions.
2. Show the essential steps.
3. Include a small example that can be checked.

This PDF accompanies the sample blog article *From equations to documents*.